

ISO 9001:2008 Certified Institute

JAVA INSTITUTE FOR ADVANCED TECHNOLOGY



PROFESSIONAL DIPLOMA IN SOFTWARE ENGINEERING

WEB PROGRAMMING I

H7DT

W.KAVEESHA GIMHANI
200480001209

COHORT 9 - LOAN INITIAL PAYMENT PAID

<COLOMBO BRANCH>



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1. INTRODUCTION

This company, CampusHub, helps students and runs sports programs, events, communities, and groups at many schools. Social media, texts, and spreadsheets are now what employees use to keep track of everything. This takes a long time and makes mistakes.

This essay talks about how to plan and create a web application for CampusHub. The system lets students read messages, look at pictures and videos, make their own profiles, and sign up for events online. A separate place for administrators to handle students, events, and content.

PHP, MySQL, HTML5, and CSS3 were used to make this game. When it comes to multimedia, HTML5 also has music and video players, and XML is used for a structured data feed.

1.1 Assumptions

The following ideas were considered in the building of the system:

- The system runs on an Apache, MySQL and PHP 8.x server (LAMP/WAMP)
- One supervisor login. The login file specifies the login info.
- All events are open to students from any school unless stated otherwise.
- Your uploaded files are stored in the /uploads/ folder at the server.
- The administrator uploads audio and video materials directly to the site.

This project is not meant to send emails as notifications.

2. Requirements Analysis

2.1 Type Of Users

User Type	Who They Are	Access Level
Student	Enrolled students who use the portal to join events	Normal User
Administrator	CampusHub Staff who manage all data and content	Full Access
Visitor	Anyone Who visits the site without login in	Read Only

2.2 What Students Need To Do

- Register an account and log in.
- Check out a calendar of forthcoming events.
- Please complete the online form to register for each event.
- See and edit their own profile page.
- Embed an image into your page.
- Browse the photo gallery and upload images to the website.
- Leave a comment with the contact form.
- See what the administrator says.
- Watch a movie on campus and listen to some music

2.3 What Administrator need to do

• Log in at a secure admin location.

You can create, update or delete events. And pictures from those events.

• View the list of students registered for each event.

• Review and delete students' accounts.

You can post and remove messages.

• Review the summary data on the admin dashboard.

2.4 Other Requirement

- Passwords should only be saved in encrypted hashes for security reasons. All DB searches should be performed using prepared statements.
- Easy to use: The website must be easy to use on PCs and phones
- Reliable: All forms should validate the information you submit and provide clear error notifications.
- Easy to maintain: The code has reusable files like header, footer, and database link.

3. Website Planning

3.1 Purpose OF THE System

Students should be able to identify events and sign up for them on the main CampusHub site. The system uses a live database rather than manual files. Administrators can submit events and communications instantly and students obtain information instantly.

3.2 Page List and Navigation

Page Name	File	Who can see it
Home page	index.php	Everyone
Event List	php/events.php	Everyone
Event Registration	php/register_event.php	Everyone
Photo gallery	php/gallery.php	Everyone
Contact Us	php/contact.php	Everyone
Sign Up	php/signup.php	Everyone
Login	php/login.php	Everyone
My Profile	php/profile.php	Logged-in students
Admin Dashboard	php/admin/dashboard.php	Admin only
Manage Events	php/admin/manage_events.php	Admin only
Manage Students	php/admin/manage_students.php	Admin only
Registrations	php/admin/manage_registrations.php	Admin only
Announcements	php/admin/manage_announcements.php	Admin only

3.3 Benefits of the New System

• All data is stored in a database so you don't have to use handwritten spreadsheets.
• Students can sign up for events anytime, anywhere.
• Administrators can make content updates quickly without IT involvement.
• All media information and event pictures are stored in one place.
• The system can be easily expanded when new events and students are added.

4. User Interface Design

4.1 Design choices

• The header and footer are the same on all pages and are created with PHP includes.
• Blue (#2563eb) is the main colour of the navigation bar and headlines.
• Use Amber/Gold (#f59e0b) to make buttons stand out.
• Card-style boxes for events and announcements that looks neat.
• The design instantly adjusts to tablets and smartphones.
• Forms provide clear labels, highlighted input fields and error notifications.

4.2 Color Scheme

Color Name	Hex Code	Where It Is Used
Blue(primary)	#2563eb	Navigation, headings, buttons, table headers
Dark Navy	#1e3a8a	Hero banner gradient, admin sidebar
Amber / Gold	#f59e0b	Call-to-action buttons, logo accent
Dark Text	#1e293b	Body text, sidebar background
Gray Text	#64748b	Small labels, date text, captions
Light Background	#f1f5f9	Page background, form fields, code blocks

4.3 Page Layout Description

Home Page

On top there is a big blue sign with two buttons and a welcome message. Next to the announcement cards are event cards in a three column grid. Then a multimedia area with neighboring audio and video players.

Events Page

At the top there is a search box and a category drop-down. Event scores are shown as cards in a grid. Each card shows an image of the event, a category badge, a title, a date, a location and a Register button.

Event Registration Form,

A simple, one column form asking for the teacher, event selection, phone number, email address, and full name. If any fields are invalid, errors are shown at the top of the form.

Admin's Dashboard

Other pages are linked in the sidebar on the left. The fours are on the right side.

5. HTML and CSS Development

5.1 HTML5 Elements Used

The site employs HTML5 semantic features to make the page structure clear and readable

HTML Element	Where It Is Used
<header>	Contains the logo and navigation menu on every page
<nav>	Navigation links changes based on whether user is logged in
<main>	The main content area of each page
<section>	Groups related content (hero, events list, media section)
<footer>	Three-column footer with links and contact information
<form>	All data entry forms (registration, login, contact, upload)
<table>	Admin data tables with thead and tbody
<audio> / <video>	Embedded players on the home page
<input> / <select>	Inputs and dropdown menus for forms

5.2 External CSS File

There is one file, css/style.css, with all the styles. This file is referenced in the header, and applies to all pages. No inline styles exist except from some minor layout tweaks.

5.3 CSS Features Used

CSS Variables (Custom Properties)

Variables are used for holding colours and common values at the front of the CSS file. This makes it very easy to change the colour scheme in one place:

```
:root { --primary: #2563eb; /* main blue */ --secondary: #f59e0b; /* gold for buttons */ --dark: #1e293b; /* dark text */ --radius: 8px; /* corner rounding */ --shadow: 0 2px 12px rgba(0, 0, 0, 0.08); }
```

Types of Selectors Used

- Element selectors : body, h1, h2, a, img, ul, table
- Class selectors include event-card, btn-primary, form-group, and admin-panel.
- Pseudo classes: :focus (form inputs), :hover (buttons and cards), :last child
- Child selectors: event-card:hover td, main-nav ul
- Attribute selectors: input[type="file"]

Design Flexibility

Gallery images & event cards use CSS Grid with repeat(auto-fit, minmax()) On a phone, the grid automatically changes from three columns to one column on a PC. You also have a media query at 768 pixels that switches the admin panel layout from side-by-side to stacked.

6. PHP Development

6.1 Variables

PHP variables store form data, database results and session data. All text input is cleaned with trim() and cast to the correct type before use:

```
$name = trim($_POST["name"] ?? ""); // text $email = trim($_POST["email"]?:"");"); // $text = $_POST["event_id"]?? 0; // $sel_id = (int) $text; // integer $success = $error = ""; // empty strings
```

6.2 Function Used

Function	File	What It Does
password_hash()	signup.php	Encrypts the password before saving
password_verify()	login.php	Checks password at login
filter_var()	All forms	Validates email addresses
htmlspecialchars()	All output pages	Stops XSS attacks on output
trim()	All forms	Removes extra spaces from input
move_uploaded_file()	gallery.php	Saves uploaded file to disk
date()	Many pages	Formats dates for display
session_start()	All pages	Starts the user session
preg_match()	register_event.php	Checks phone number format
uniqid()	gallery.php	Creates a unique filename
pathinfo()	gallery.php	Gets the file extension

6.3 PHP Includes (Reusable Files)

In includes/ folder there are 3 files per page. This prevents the same code from being duplicated into every file:

```
include "includes/db.php"; // database connection include "includes/header.php"; // HTML head tag + navigation include "includes/footer.php"; // footer + closing HTML tags
```

6.4 Forms and Data Transfer Between Pages

Forms transfer data securely using POST . The event ID is passed across pages using GET in the URL:

```
// Link on events page passes the event ID href="register_event.php?id=3" // register_event.php reads it $event_id = isset($_GET["id"]) ? (int)$_GET["id"] : 0;
```

6.5 Session management

PHP sessions will keep user logged in when navigating between pages. On secure pages://

Check whether set on protected pages

```
// Set at login $_SESSION["user_id"] = $user["id"]; $_SESSION["user_name"] =
$user["fullname"]; // Check on protected page if (!isset($_SESSION["user_id"])) {
header("Location: login.php"); exit; }
```

7. Database Connectivity and Data Manipulation

7.1 Database Tables

The database is named campushub_db and has six tables:

Table Name	What It Stores	Important Columns
students	Student accounts	id, email, password, faculty
events	Campus events	id, title, event_date, category
registrations	Who signed up for what	id, event_id (link to events), email
announcements	Admin notices	id, title, content
gallery	Photo records	id, filename, caption
contact_messages	Contact form messages	id, email, subject, message

7.2 Connecting to the Database

$conn = new mysqli("localhost", "root", "", "campushub_db"); if ($conn->connect_error) { die("Cannot connect: " . $conn->connect_error); } $conn->set_charset("utf8");

7.3 INSERT – Adding a Registration

$stmt = $conn->prepare("INSERT INTO registrations (name, email, phone, event_id, faculty, registered_at) VALUES (?, ?, ?, ?, ?, NOW())"); $stmt->bind_param("sssiss", $name, $email, $phone, $event_id, $faculty); $stmt->execute();

7.4 SELECT – Getting Upcoming Events

$sql = "SELECT * FROM events WHERE event_date >= CURDATE()"; if ($search !== "") { $sql .= " AND title LIKE ?"; } $stmt = $conn->prepare($sql); $stmt->execute(); $events = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);

7.5 UPDATE – Editing a Student Profile

$stmt = $conn->prepare("UPDATE students SET fullname=?, phone=?, faculty=? WHERE id=?"); $stmt->bind_param("sssi", $fullname, $phone, $faculty, $user_id); $stmt->execute();

7.6 DELETE – Admin Removes an Event

$id = (int)$_GET["delete"]; // cast to integer for safety $conn->query("DELETE FROM events

WHERE id=$id");

7.7 Security

• Prepared statements are utilised for every INSERT, UPDATE and SELECT queries. This prevents SQL injection attacks.
• Passwords are encrypted with the bcrypt encryption method, using the password_hash() function.
• All text output to the screen is routed through htmlspecialchars() to prevent script injection.

8. XML Implementation

The event data is structured in an XML file (xml/events.xml). This can be leveraged by mobile apps and other systems that need to read CampusHub events. The rules for the XML document are defined in a DTD file (xml/events.dtd).

8.1 XML Structure Files

Basketball Tournament Sports 07/15/2026 There is an annual inter-faculty competition in the main sports complex. Committee for Sports

8.2 Document Type Definition (DTD)

The DTD file checks that the structure of the XML is correct:

<!ELEMENT campushub (events)>
<!ELEMENT events (event+)>
<!ELEMENT event (title, category, date, location, description, organiser)>
<!ATTLIST event id CDATA #REQUIRED>
<!ELEMENT title (#PCDATA)>
<!ELEMENT category (#PCDATA)>

8.3 Why Use XML

• Other programs, such mobile apps, can read XML.
• It contains information in an orderly, readable fashion.
• The DTD guarantees that all the mandatory fields are included for each event record.
• It shows an alternative way of exchanging and storing data outside databases.

9. Multimedia and File Upload Features

9.1 Video and Audio Player

The home page features an HTML5 video player and an audio player in the Campus Life section. Both have a fallback message for ancient browsers:

<video controls width="100%">
<source src="uploads/campus_intro.mp4" type="video/mp4">

Your browser cannot play this video.
</video>
<audio controls style="width:100%">
<source src="uploads/campus_anthem.mp3" type="audio/mp3">
Your browser cannot play this audio.
</audio>

9.2 Gallery Photo Upload

The gallery site allows anyone to upload a photo. The system verifies the file prior to saving it:

Check	Rule
File type	Only JPEG, PNG, GIF or WebP are allowed
File size	Maximum 5 MB
PHP error	Upload error code is checked first

How the File is Saved

// Get file extension $ext = pathinfo($file["name"], PATHINFO_EXTENSION); // Create a unique filename to avoid conflicts $filename = "gallery_" . uniqid() . "." . strtolower($ext); $dest = dirname(__DIR__) . "/uploads/" . $filename; // Move to uploads folder and save record if (move_uploaded_file($file["tmp_name"], $dest)) { $stmt->execute(); // save filename in database }

9.3 Profile Picture Upload

Students also have the ability to upload a photo to their personal page. Size and type tests are the same. The file is avatar_{user_id}.jpg so it automatically replaces the previous file.

10. Error Handling and Validation

10.1 Form Validation Rules

Field	Rule	Page
Name	Must be at least 3 characters	Sign up, Registration
Email	Must be a valid email format	All forms
Phone	Numbers only, 7 to 15 digits	Registration
Password	At least 8 characters; must match confirm	Sign up
Event selection	Must pick an event (cannot leave blank)	Registration
File type	Must be JPEG, PNG, GIF or WebP	Gallery, Profile

File size	Must not exceed 5 MB (2 MB for avatars)	Gallery, Profile

10.2 How Errors are shown

All validation errors are collected in an array and merged into one message displayed above the form in a red box:

$errors = []; if (empty($name)) $errors[] = "Name is required."; if (!filter_var($email, FILTER_VALIDATE_EMAIL)) $errors[] = "Please enter a valid email."; if (!empty($errors)) { $error = implode(" | ", $errors); }

10.3 File Upload Errors

if ($file["error"] !== UPLOAD_ERR_OK) { $error = "Upload failed. Please try again."; } elseif (!in_array($file["type"], $allowed)) { $error = "Only JPEG, PNG, GIF and WebP are allowed."; } elseif ($file["size"] > $max_size) { $error = "File must be smaller than 5 MB."; }

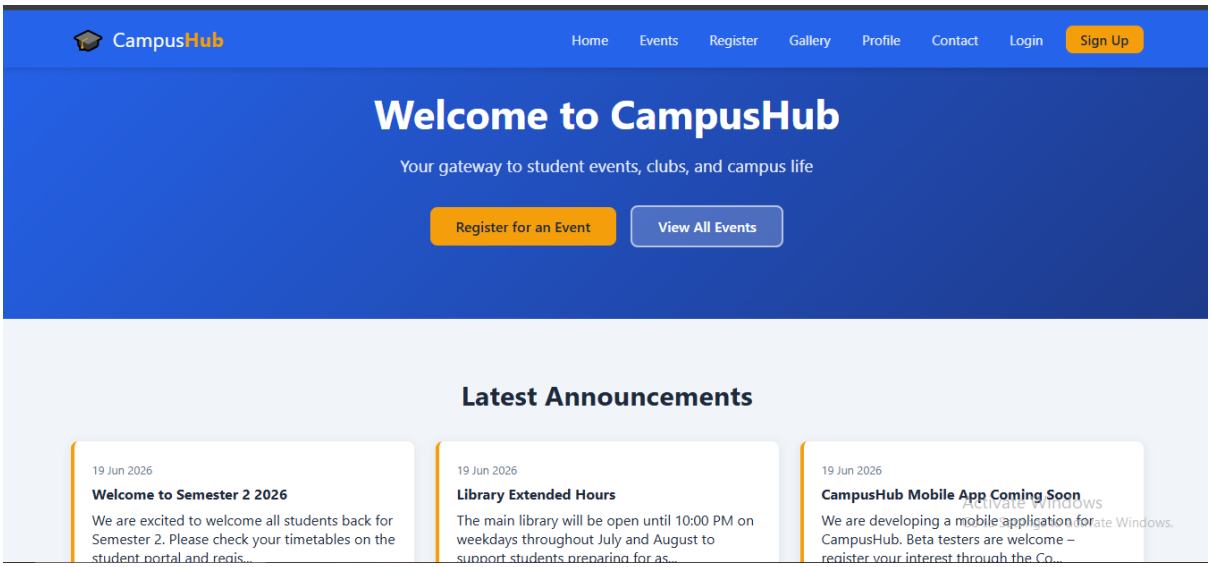
10.4 Page Access Control

If the user is not logged in, protected sites redirect to the login page:

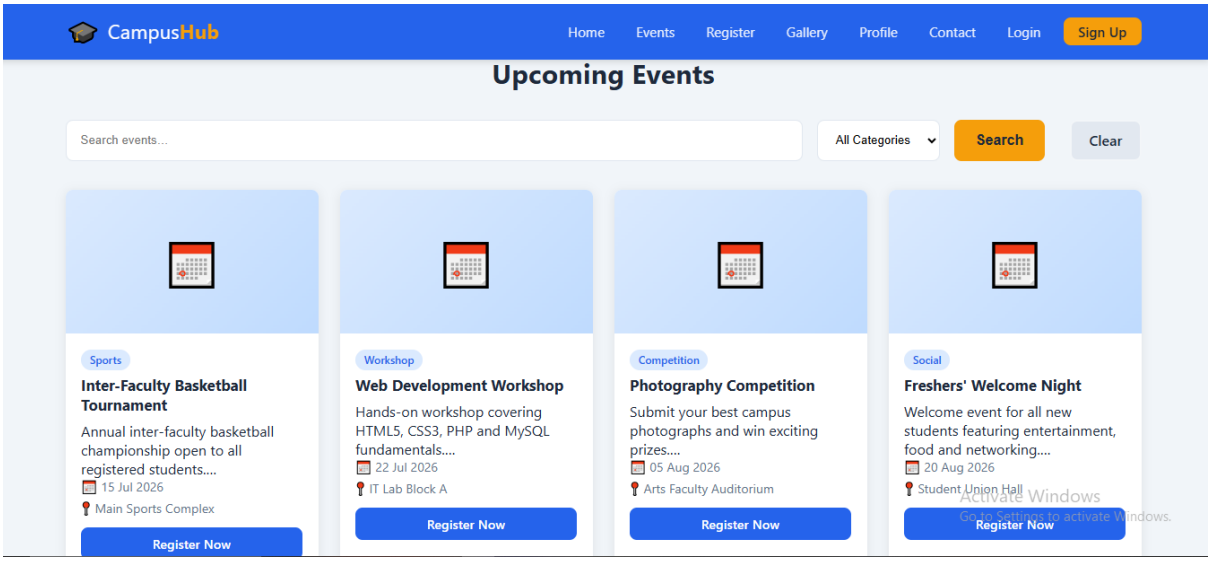
// Student page protection if (!isset($_SESSION["user_id"])) { header("Location: login.php"); exit; } // Admin page protection if (!isset($_SESSION["admin"])) { header("Location: /campushub/php/login.php"); exit; }

11.ScreenShots of the Web Application

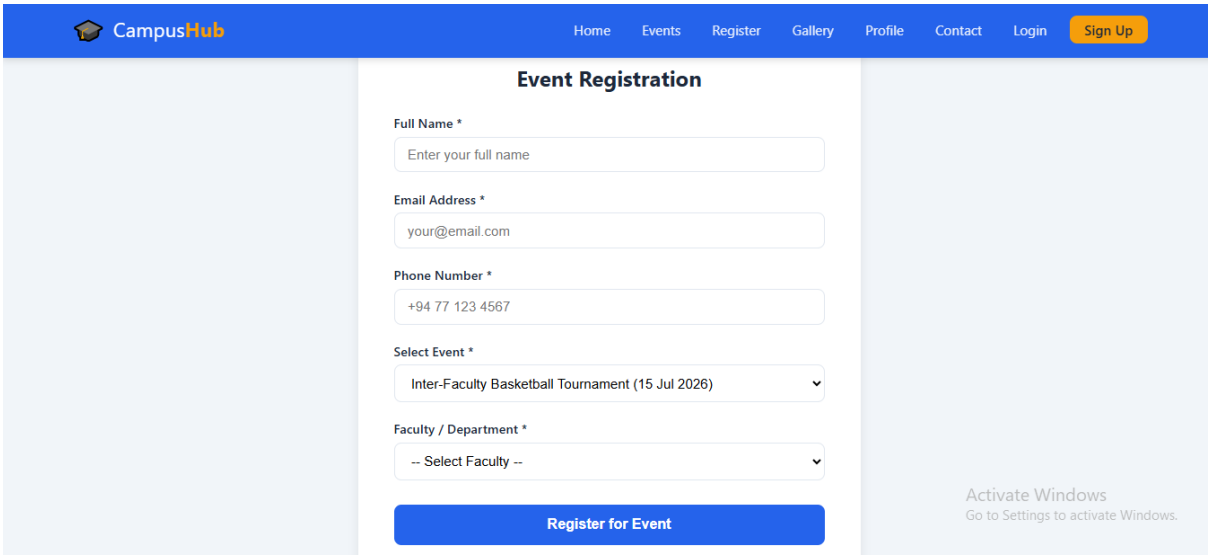
11.1 Home Page



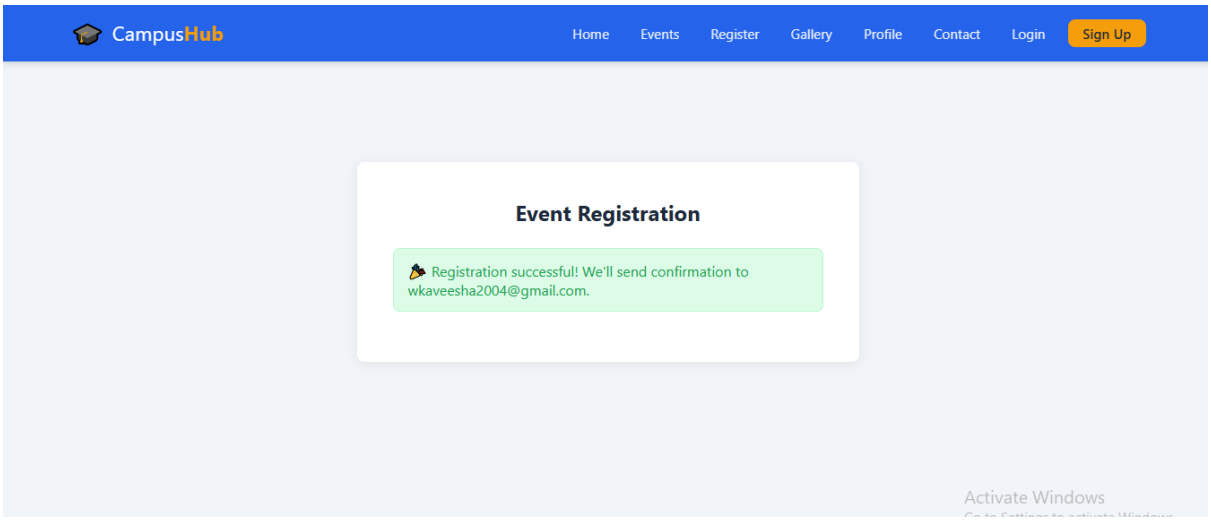
11.2 Event Listing Page



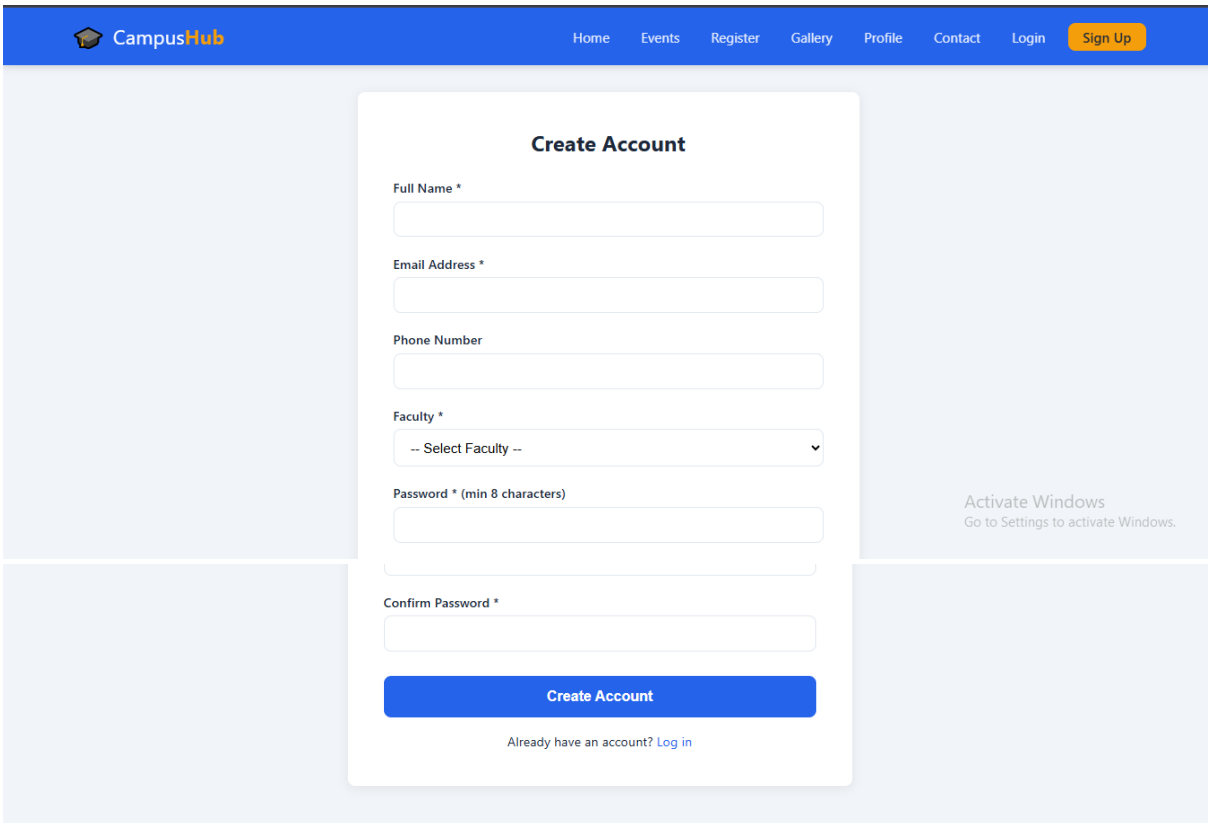
11.3 Event Registration Form



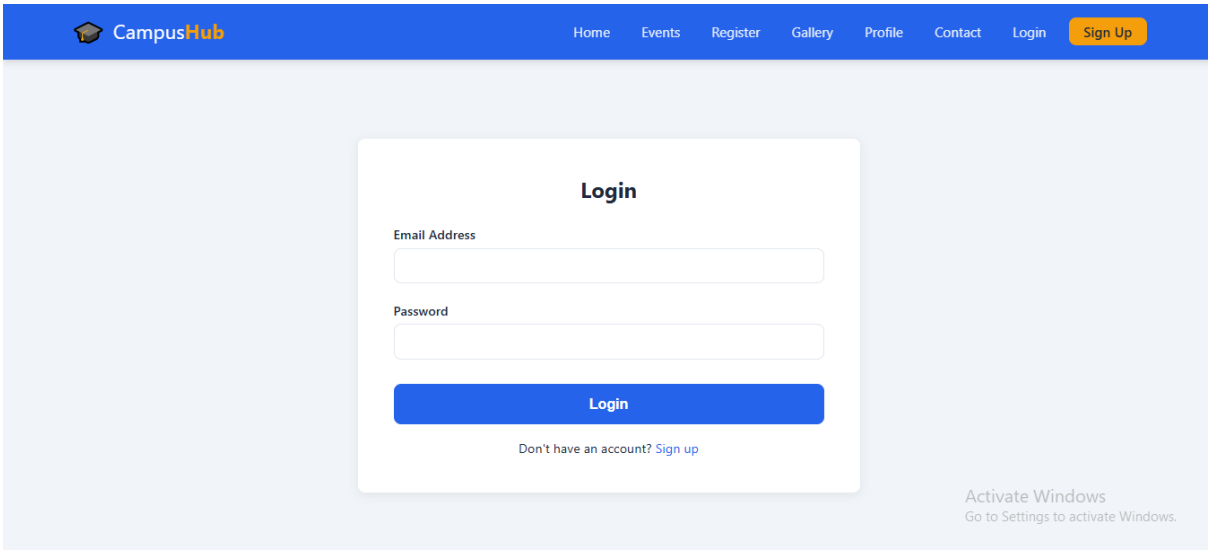
11.4 Registration Success Message



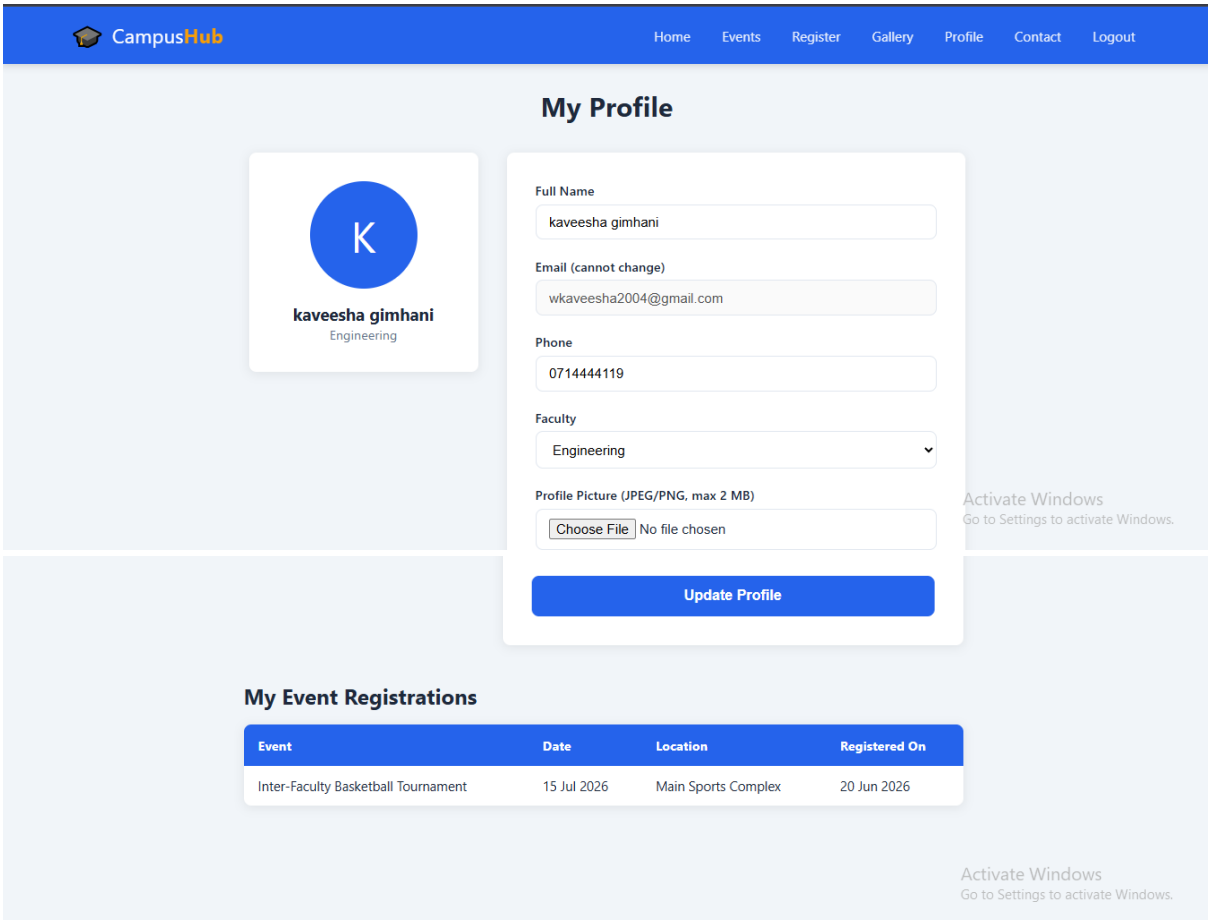
11.5 Sign up page



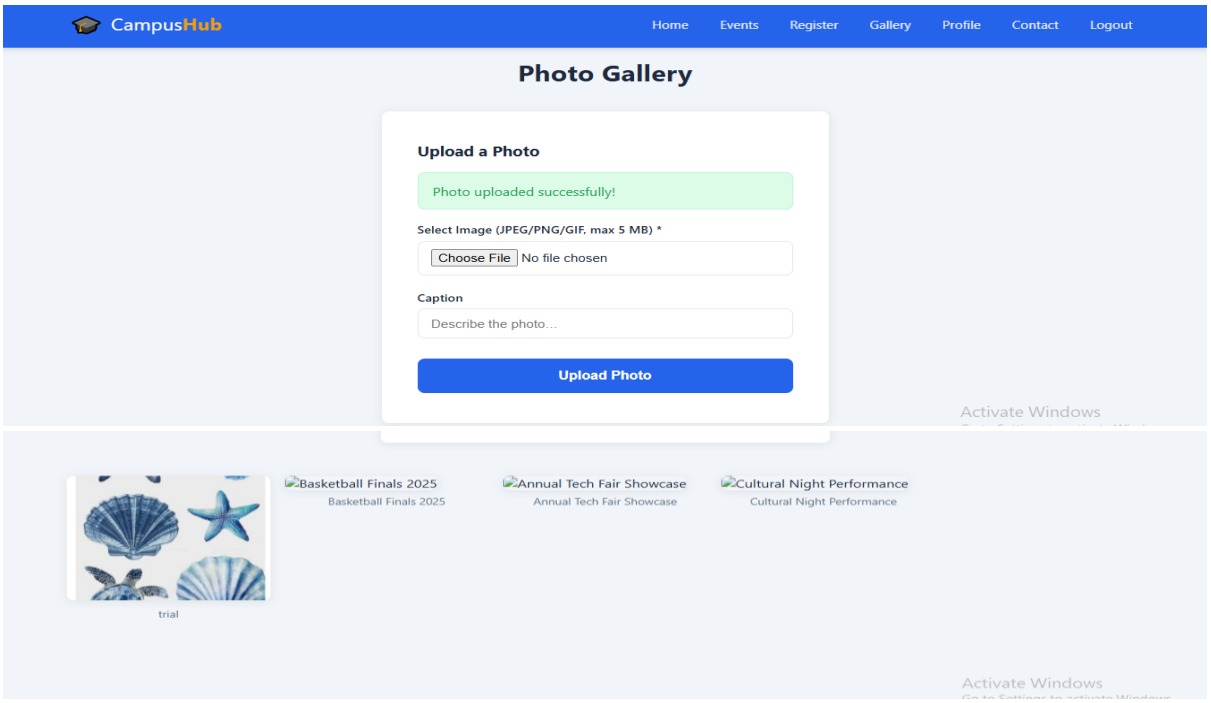
11.6 Log in page



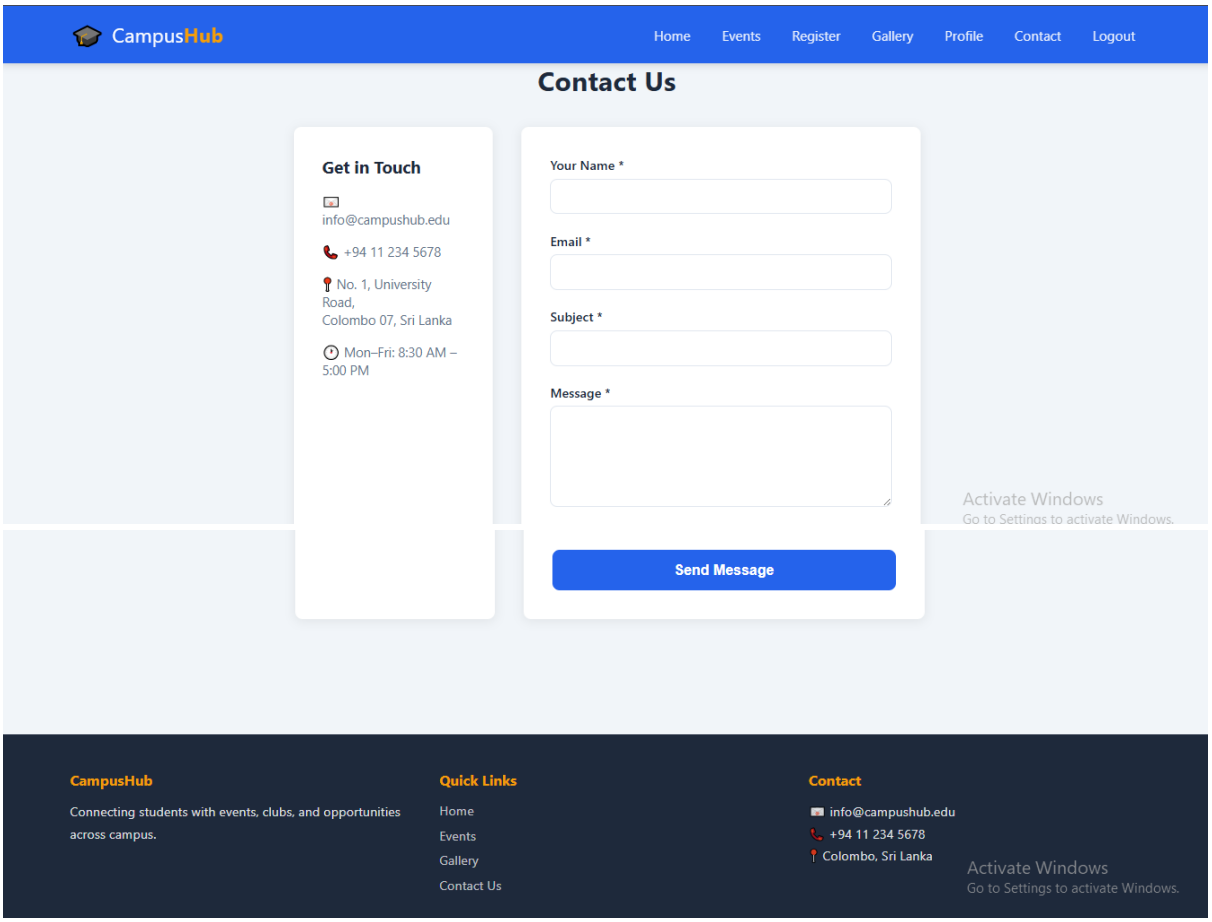
11.7 Student Profile Page



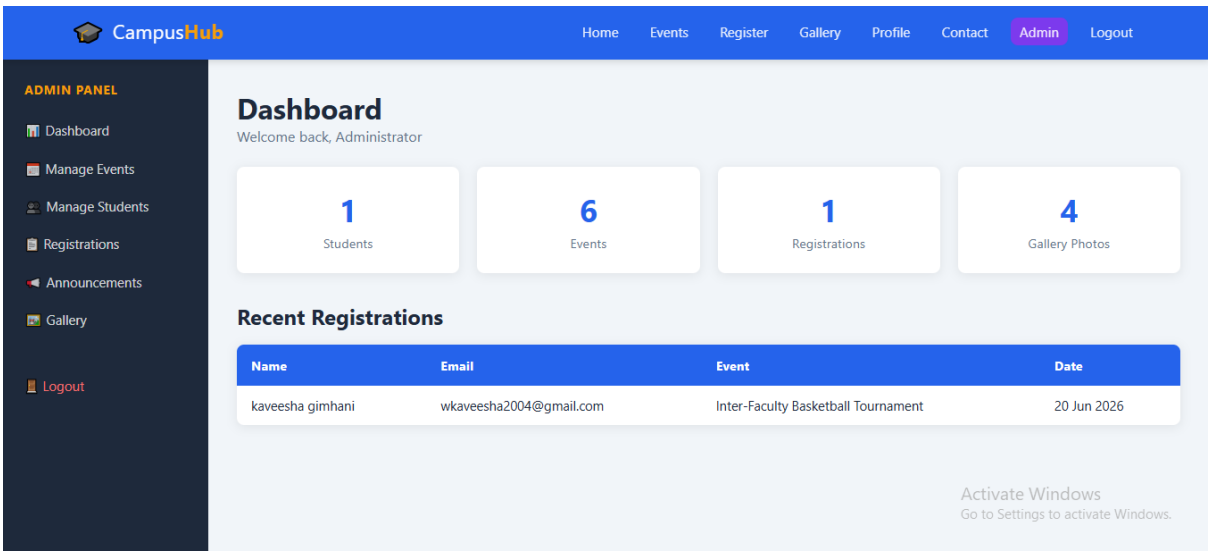
11.8 Photo Gallery



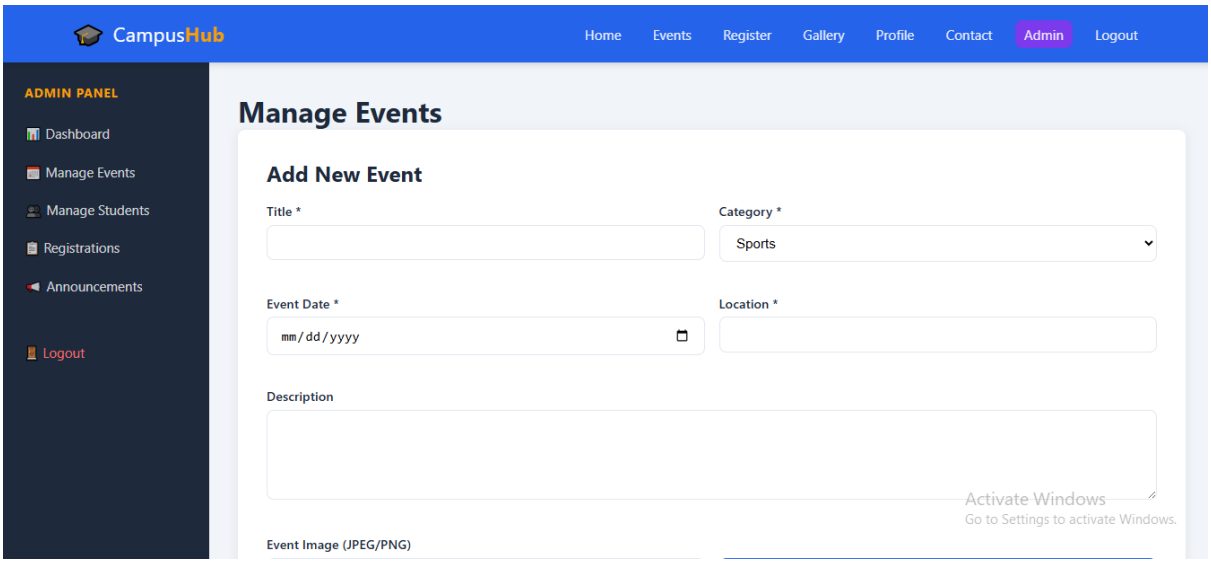
11.9 Contact Page



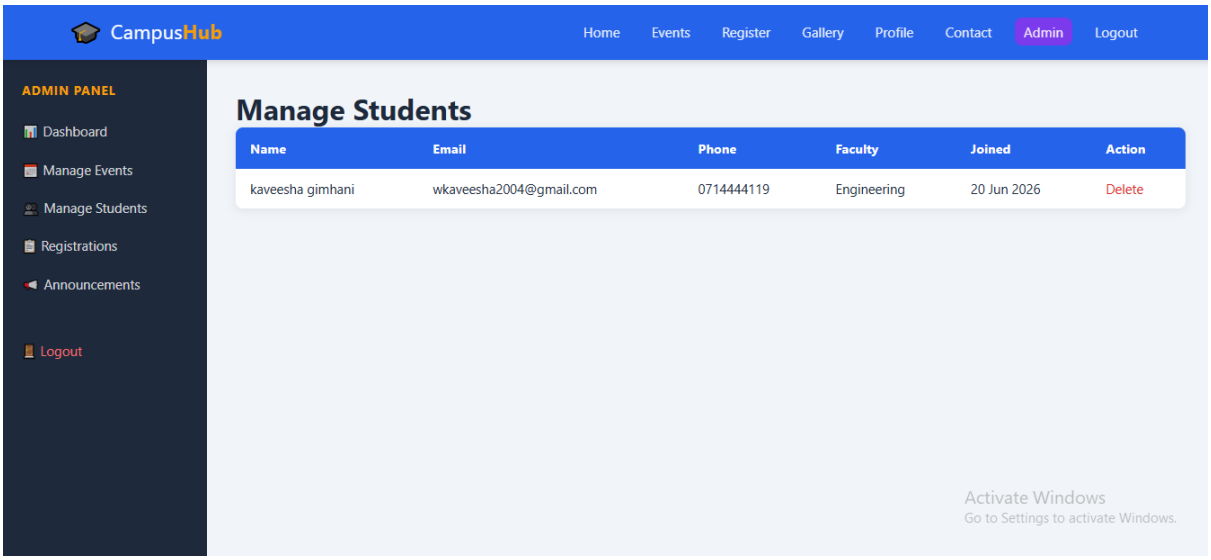
11.10 Admin Dashboard



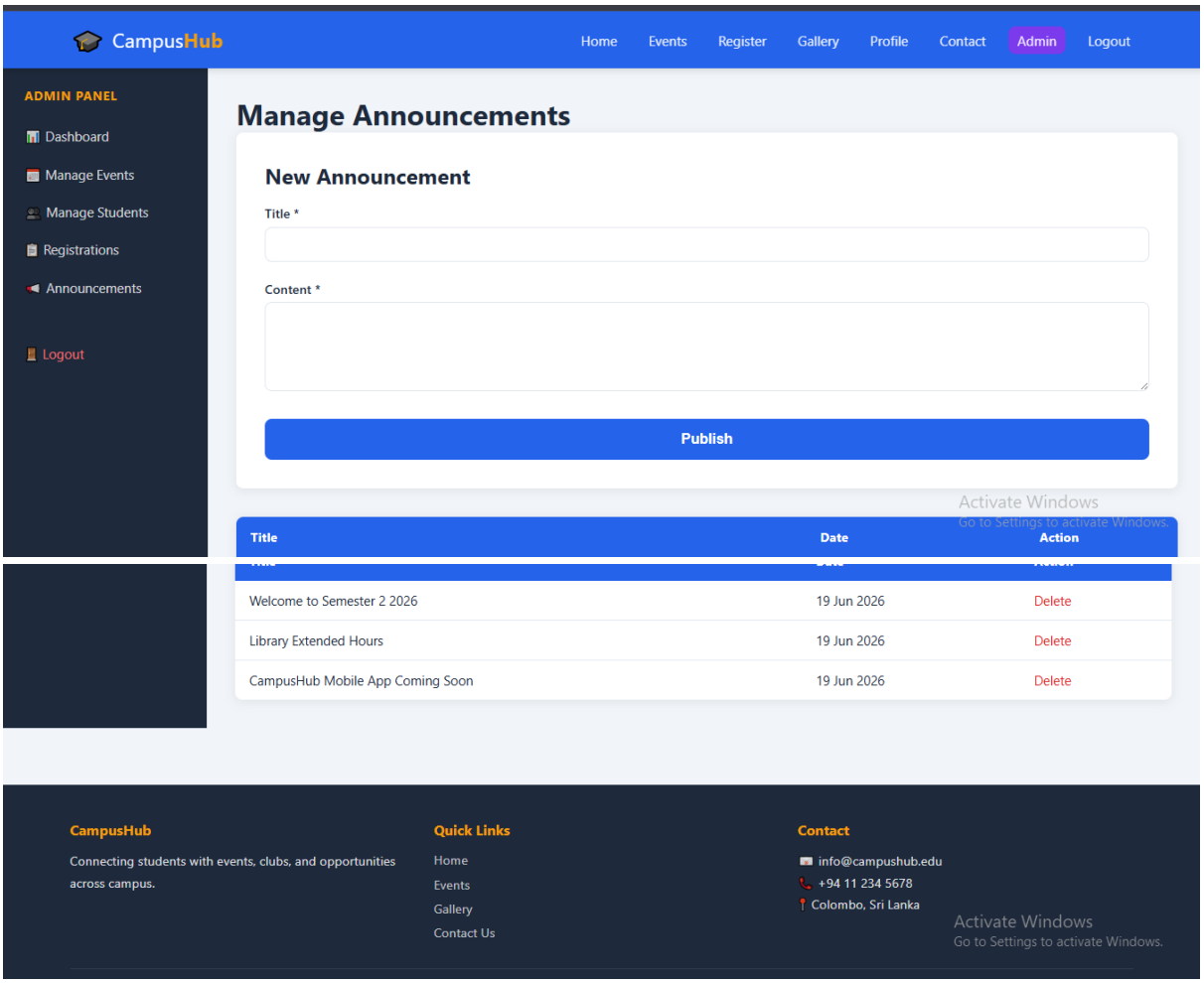
11.11 Admin - Manage Events



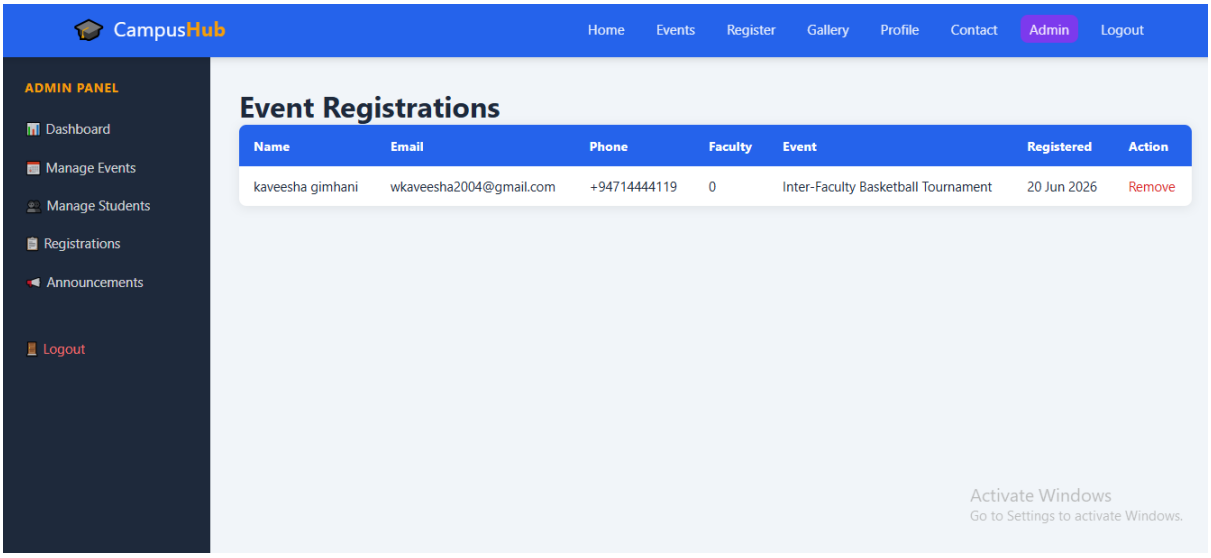
11.12 Admin – Manage Students



11.13 Announcements



11.14 Admin – Registration List



12.Conclusion

The CampusHub web application has been designed and built successfully. It replaces tedious manual processes with a fast, easy to use website that students and administrators may use from any device.

What the system does (mostly)

- Students can register for events online through a fully validated form.
- Admins can add, update and delete events & announcements from admin panel.
- Photo gallery for uploading and viewing campus images.
- HTML5 audio and video players are used to showcase multimedia content on the home page.
- Student accounts are safeguarded by encrypted passwords and session management.
- DTD validated XML file with structured data stream.
- The site uses one external CSS file with variables to keep the design constant.
- All database queries are done using prepared statements to prevent SQL injection.

The system can be improved in the future by adding an email confirmation after registration, a student attendance tracker and a mobile app that consumes the XML data feed.

Appendices

Project File Structure

campushub/
 index.php Home page
campushub_db.sql Database script
 css/
style.css Main stylesheet (external CSS)
includes/

db.php Database connection
header.php Site header and navigation
footer.php Site footer
php/
events.php Events listing page
register_event.php Event registration form
gallery.php Photo gallery and upload
profile.php Student profile
signup.php New account registration
login.php Login page
logout.php Destroys session and logs out
contact.php Contact form
admin/
dashboard.php Admin overview and stats
manage_events.php Add, edit, delete events
manage_students.php View and delete students
manage_registrations.php View and delete registrations
manage_announcements.php Publish and delete announcements
xml/
events.xml XML events data feed
events.dtd DTD validation rules
uploads/ All uploaded images and media

Database Schema

Table	Column	Type & Notes
students	id	INT, PK, AUTO - Primary key
students	fullname	VARCHAR(150)
students	email	VARCHAR(150) - Must be unique
students	password	VARCHAR(255) - Bcrypt hash
students	phone	VARCHAR(20)
students	faculty	VARCHAR(100)
students	avatar	VARCHAR(255) - Profile picture filename
events	id	INT, PK, AUTO - Primary key

events	title	VARCHAR(200)
events	event_date	DATE
events	location	VARCHAR(200)
events	category	VARCHAR(80)
events	image	VARCHAR(255) - Event image filename
registrations	id	INT, PK, AUTO - Primary key
registrations	event_id	INT, FK - Links to events.id
registrations	name	VARCHAR(150)
registrations	email	VARCHAR(150)
gallery	id	INT, PK, AUTO - Primary key
gallery	filename	VARCHAR(255) - Saved file name
gallery	caption	VARCHAR(300)

Admin Log In Credinals

Role	Email	Password
Administrator	admin@campushub.edu	Admin@1234